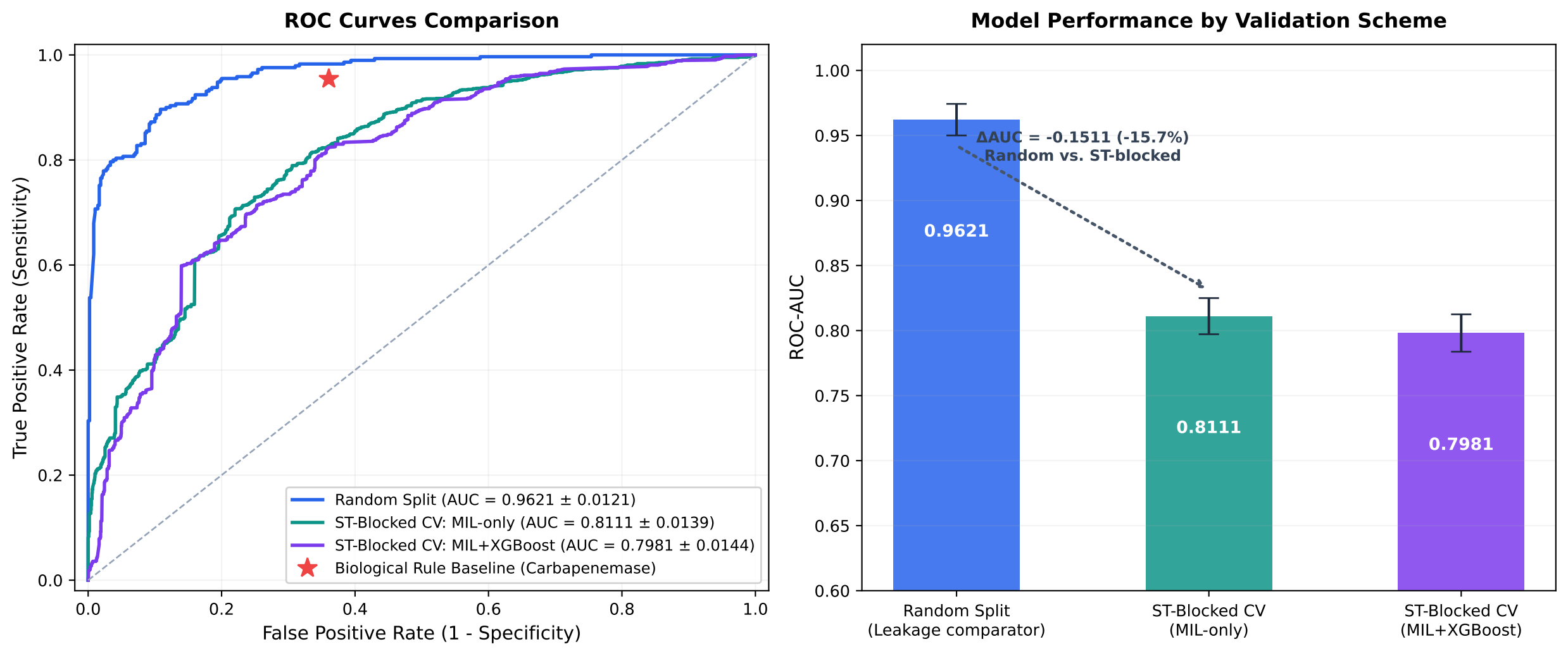
ROC Curves Comparison
True Positive Rate (Sensitivity)
False Positive Rate (1 - Specificity)
0.0
0.2
0.4
0.6
0.8
1.0
Random Split (AUC = 0.9621 ± 0.0121)
ST-Blocked CV: MIL-only (AUC = 0.8111 ± 0.0139)
ST-Blocked CV: MIL+XGBoost (AUC = 0.7981 ± 0.0144)
Biological Rule Baseline (Carbapenemase)
Model Performance by Validation Scheme
ROC-AUC
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
ΔAUC = -0.1511 (-15.7%)
Random vs. ST-blocked
0.9621
0.8111
0.7981
Random Split
(Leakage comparator)
ST-Blocked CV
(MIL-only)
ST-Blocked CV
(MIL+XGBoost)